

Uniqueness among all state extensions of the CAR trace

MathlibAnnex.CStarAlgebra.CAR.existsUnique_state_extension_trace

theorem

Pointed transport identifies arbitrary state extensions before traciality is proved.

Statement

Let A be the completed CAR algebra and τ its normalized trace. Fix a representative shell family and its chosen unitary links L_j in the selected atomic representation π . Write $T = C^*(\pi(A), \{L_j\})$ for the resulting concrete target and $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, for its faithful unital source map. This fixes one target and one source map throughout. There is exactly one state ϕ on T such that $\phi \circ \iota = \tau$. Uniqueness here ranges over all state extensions, including those not assumed tracial.

Assumptions

Only the fixed representative shell family is given. A competing functional ψ must be a state on T and satisfy $\psi(\iota(a)) = \tau(a)$ for every $a \in A$.

Conclusion

Every such ψ equals the chosen extension ϕ . The existence and uniqueness refer to the same fixed algebra T .

The argument does not use uniqueness among tracial states to prove this stronger extension uniqueness. The unitary below compares the GNS representations of two possibly different extension states; it is not an identification of independently chosen vectors without a transport theorem.

Proof route

Construct the two target GNS representations and apply pointed unitary transport from their common source trace. Equality of the vector functionals then gives equality of states.

▼ Proof steps

1. Existence is `exists_state_extension_trace`. For an arbitrary competing state ψ , let $(H_\psi, \rho_\psi, \xi_\psi)$ be its GNS representation; define $(H_\phi, \rho_\phi, \xi_\phi)$ for the chosen extension. Both target orbits are dense and their source vector functionals are τ .
2. Apply `exists_pointed_unitary_of_trace_of_cyclic` with the ψ representation first and the ϕ representation second. Its hypotheses are exactly the two restriction equalities and target cyclicity. It gives a unitary

$$E_\psi : H_\psi \longrightarrow H_\phi, \quad E_\psi \xi_\psi = \xi_\phi, \quad E_\psi \rho_\psi(t) = \rho_\phi(t) E_\psi \quad (t \in T).$$

3. For every $t \in T$, preservation of the inner product and these two identities give

$$\begin{aligned} \psi(t) &= \langle \xi_\psi, \rho_\psi(t) \xi_\psi \rangle \\ &= \langle E_\psi \xi_\psi, E_\psi \rho_\psi(t) \xi_\psi \rangle \\ &= \langle \xi_\phi, \rho_\phi(t) \xi_\phi \rangle = \phi(t). \end{aligned}$$

This is the pointwise equality used by

`eq_traceExtension_of_mem_stateSpace_of_apply_shellFamilySourceHom_eq_trace`; extensionality then gives $\psi = \phi$.

Main citations

- The stated existence or structural result —
`MathlibAnnex.CStarAlgebra.CAR.existsUnique_state_extension_trace`
- The fixed concrete target and source map —
`MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom`
- Faithfulness of the source embedding —
`MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom_injective`
- Existence of a state extension — `MathlibAnnex.CStarAlgebra.CAR.exists_state_extension_trace`
- Pointed transport for trace-realizing cyclic targets —
`MathlibAnnex.CStarAlgebra.CAR.exists_pointed_unitary_of_trace_of_cyclic`
- Equality of an arbitrary extension with the chosen state —
`MathlibAnnex.CStarAlgebra.CAR.eq_traceExtension_of_mem_stateSpace_of_apply_shellFamilySourceHom_eq_trace`

Supporting route explanation

Lean source signature (exact)

```
theorem existsUnique_state_extension_trace (family : RepresentativeShellFamily) :
  ∃! ϕ : ShellFamilyTarget family → L[ℂ] ℂ,
    ϕ ∈ MathlibAnnex.Analysis.CStarAlgebra.stateSpace (ShellFamilyTarget family) ∧
    ∀ b, ϕ (shellFamilySourceHom family b) = trace b
```

In the source	Mathematical meaning
<code>family : RepresentativeShellFamily;</code> <code>ShellFamilyTarget family</code>	The fixed CAR shell family and once-chosen links L_j determine the same concrete target $T = C^*(\pi(A), \{L_j\})$ throughout.
<code>shellFamilySourceHom family; trace</code>	The faithful unital source map $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, and the normalized CAR trace $\tau : A \rightarrow \mathbb{C}$.
<code>∃! ϕ : ShellFamilyTarget family → L[ℂ] ℂ</code>	There exists exactly one continuous complex-linear functional $\phi : T \rightarrow \mathbb{C}$ satisfying the two following conditions together.

In the source	Mathematical meaning
$\phi \in \dots \text{stateSpace } (\text{ShellFamilyTarget family})$	ϕ is positive and satisfies $\phi(1_T) = 1$.
$\forall b, \phi(\text{shellFamilySourceHom family } b) = \text{trace } b$	Its restriction is $\phi(\iota(b)) = \tau(b)$ for every $b \in A$. Any other state with this same restriction equals ϕ , without being assumed tracial. This does not say that all states on T coincide.

Further source notes: The linked uniqueness supplier calls the arbitrary competing state ϕ and its positive-map bundle f ; these correspond to ψ and its positive bundle in the mathematics above. Its proof-local ρ , ξ , and e are the competitor's GNS action, vector and pointed transport toward the chosen extension. Thus E_ψ records the direction of that e .

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.CAR.existsUnique_state_extension_trace
 Accepted content SHA-256: 7cc334a512bcad3c43e64c59aa327eebe3bab0ab009cb16edc4bb10ead4246c0
 Accepted source guide SHA-256: 47b788ef24df2c4709a275c1dd408f69f0c1c06a10ced2368a90d8c8cf3f452f
 Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73